Gowin FIFO IP

User Guide

IPUG105-1.08E,09/26/2025

Revision History

Date	Version	Description
08/12/2016	1.00E	Initial version published.
10/27/2016	1.01E	Applied to the GW2AR series of FPGA products.
03/01/2018	1.02E	FIFO/FIFO SC configuration information updated.
04/20/2018	1.03E	● Performance and resource of GW2AR-55 and GW2AR-18 IP updated. ● Features of the new FWFT updated.
07/17/2018	1.04E	● Performance and resource of GW2AR-55 and GW2AR-18 IP updated. ● GW1N-4 performance and resource added. ● New resource statistics approach used.
03/28/2019	1.05E	Some timing diagrams updated.
06/15/2020	1.06E	● IP configuration figures updated. ● The note in Table 5-1 updated.
02/04/2021	1.07E	The description of Chapter 4 updated.
09/26/2025	1.08E	● The IP configuration screenshots in chapter 4 Timing Description updated. ● The screenshots and related descriptions chapter 5 Gowin FIFO/FIFO SC IP Configuration updated. ● Section 1.2 Related Documents updated. ● Reference design removed.

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1 About This Guide

1.1 Purpose

This manual provides the descriptions of feature, port, timing, and reference design of Gowin FIFO IP to help you learn the function and usage of Gowin FIFO IP. The software screenshots in this manual are based on V 1.9.11.03(64-bit). As the software is subject to change without notice, some information may not remain relevant and may need to be adjusted according to the software that is in use.

1.2 Related Documents

The latest user guides are available on the Gowin website. Refer to the related documents at www.gowinsemi.com:

- DS100, GW1N series of FPGA Products Data Sheet
- DS117, GW1NR series of FPGA Products Data Sheet
- DS821, GW1NS series of FPGA Products Data Sheet
- DS861, GW1NSR series of FPGA Products Data Sheet
- DS881, GW1NSER Series of SecureFPGA Products Datasheet
- DS891, GW1NRF series of Bluetooth FPGA Products Data Sheet
- DS841, GW1NZ series of FPGA Products Data Sheet
- DS102, GW2A series of FPGA Products Data Sheet
- DS226, GW2AR series of FPGA Products Data Sheet
- DS971, GW2AN-18X &9X Data Sheet
- DS976, GW2AN-55 Data Sheet
- DS961, GW2ANR series of FPGA Products Data Sheet
- SUG100, Gowin Software User Guide
- DS981, GW5AT series of FPGA Products Data Sheet

- DS1111, GW5AT series of FPGA Products (Automotive) Data Sheet
- DS1103, GW5A series of FPGA Products Data Sheet
- DS1113, GW5A series of FPGA Products (Automotive) Data Sheet
- DS1239, GW5AST series of FPGA Products Data Sheet
- DS1105, GW5AS series of FPGA Products Data Sheet
- DS1108, GW5AR series of FPGA Products Data Sheet
- DS1118, GW5ART series of FPGA Products Data Sheet
- SUG100, Gowin Software User Guide

1.3 Abbreviations and Terminology

The abbreviations and terminology used in this manual are as shown in Table 1-1.

Table 1-1 Abbreviations and Terminology

Abbreviations and Terminology	Full Name/Meaning
ECC	Error Correcting Code
FIFO	First Input First Output
GSR	Global System Reset
IP	Intellectual Property
LUT	Look-up Tables
RAM	Random Access Memory

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using any of the information provided below.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 FIFO Overview

2.1 FIFO

FIFO is the abbreviation of first in first out. It is a first-in-first-out queue. Peripheral control logic controls reading from or writing to FIFO. FIFO also provides many handshake signals that interact with the peripheral control logic.

When FIFO is not full and write-enable is active, data is written to the FIFO queue at the rising edge of the CLK. A full flag indicates that FIFO is full and no more write operations will be executed.

When FIFO is not empty and read-enable is active, data is read from the FIFO queue at the rising edge of the CLK. An empty flag indicates FIFO is empty and no more read operations will be executed.

Illegal requests will not affect FIFO. If a read request is initiated when the FIFO is empty or a write request is initiated when the FIFO is full, they will not have any influence on the data in FIFO. The operations will be ignored, and the overflow and underflow flags will be set. You can determine whether the operation is illegal or not by monitoring these two signals. The water level signal will indicate how much effective data FIFO directly contains. You can also customize empty or full flags by setting different thresholds.

FIFO contains both synchronous FIFO (FIFO SC) and asynchronous FIFO (FIFO). The structures of these are shown in Figure 2-1 and Figure 2-2 respectively.

Figure 2-1 FIFO Structure

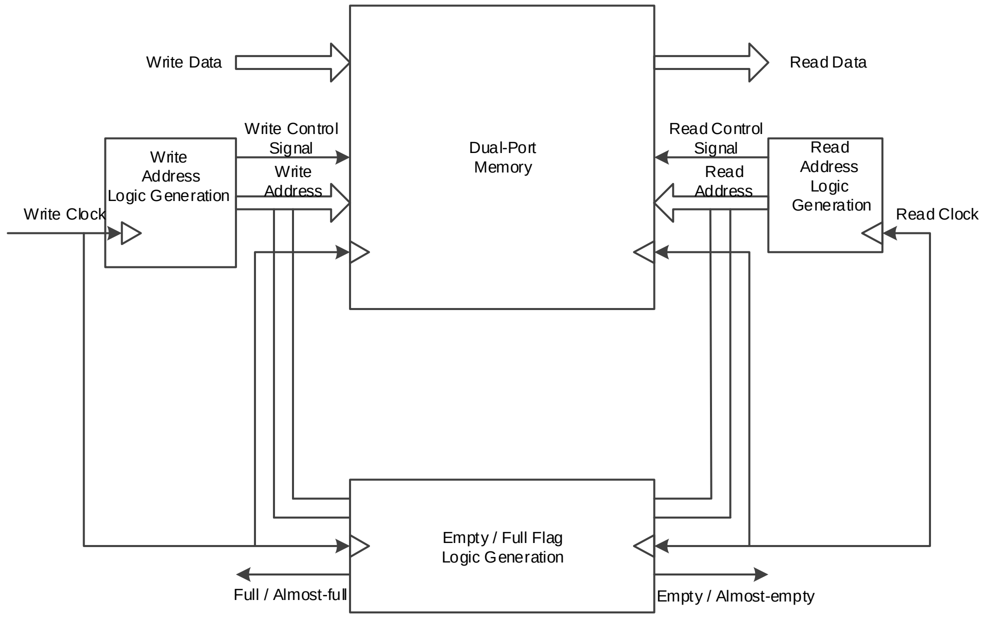
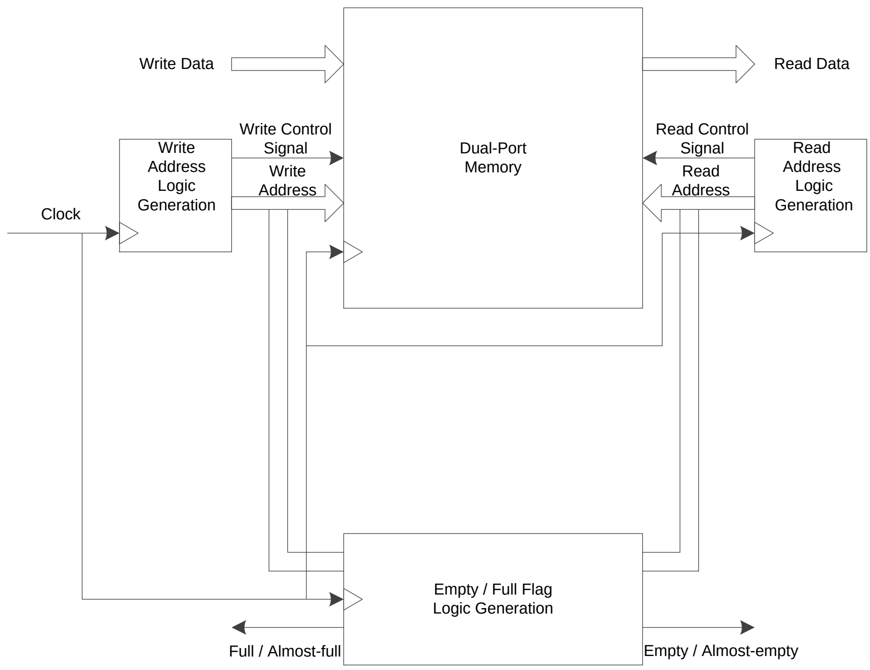


Figure 2-2 FIFO SC Structure



2.2 Gowin FIFO/FIFO SC IP Introduction

Gowin FIFO IP contains synchronous FIFO (FIFO SC) IP and asynchronous FIFO (FIFO) IP.

- The read port signal and write port signal of the synchronous FIFO are all controlled by one clock domain.
- The read port signal and write port signal of the asynchronous FIFO

are controlled by two independent clock domains.

Gowin IP Core Generator can be used to generate a FIFO IP with single-cycle read/write and configurable ports. The configuration port is such that you can generate FIFO IP of different bitwidths and data depths according to your needs.

Table 2-1 Gowin FIFO/ FIFO SC IP Overview

Gowin FIFO and FIFO SC IP	
Logic Resource	Different resources for different configurations,bit widths, and data depths.
Delivered Doc.	
Design Files	Verilog (encrypted)
Test and Design Flow	
Integrated Software	GowinSynthesis
Application Software	Gowin Software

Note!

For the devices supported, you can click here to get the information.

2.3 Gowin FIFO/FIFO SC IP Function and Feature

2.3.1 Gowin FIFO IP Function and Feature

Gowin asynchronous FIFO IP can transfer and cache the data of different bitwidths in asynchronous clock domains and you can configure different output control signals and data structures according to your requirements.

The main features of the FIFO IP are as follows:

- The implementation type of the internal memory of asynchronous FIFO is configurable, including block SRAM, shadow SRAM, and LUT.
- Write data depth is configurable. The depth is 2^n, and the maximum value is 65536.
- Write data bitwidth is configurable. The size is 1-256 bit.
- Read data depth is configurable. The depth is 2^n, and the maximum value is 65536.
- Read data bitwidth=Write data depth x Write bitwidth/Read data depth, not configurable.

  Note!

  The figures applied to the formula described above must be divisible, which limits the maximum read data depth.
- The output of the read/write data number is configurable. You can choose whether to output the read/write number data.

- The reset function is configurable. You can choose not to use reset (reset by GSR), to use one reset, or to reset read and write respectively.
- The flag signal output is optional. You can choose to output almost empty flag, almost full flag, or no flag.
- If you choose to output a almost empty or almost flag, the threshold of almost empty, almost full is configurable. They can be set as static single constant threshold, static double constant threshold, dynamic single input threshold, dynamic double input threshold.
- The ECC check function is optional. The ECC check function can be selected when the asynchronous FIFO internal memory is achieved by block SRAM and the read/write bitwidth are equal and are all below 64 bit.
- The output register function is configurable. If you choose the output register function, the read-enable (RdEn) control is optional. You choose enable-control, the output register is controlled by RdEn, and the last data cannot be output. If you choose the output register function instead of read-enable control, the read data output will be a cycle later, and the last data will be output.
- Supports first-word fall-through function.

2.3.2 Gowin FIFO SC IP Function and Feature

Gowin FIFO SC IP can transfer and cache the data of the same bitwidth and depth in the same clock domains. It can also configure and output different control signals according to the your requirements.

The main features are as follows:

- The implementation type of the internal memory of FIFO SC is configurable, including block SRAM, shadow SRAM, and LUT.
- Write data depth is configurable. The depth is 2^n, and the maximum value is 65536.
- Write data bitwidth is configurable. The size is 1-256 bit.
- The depth and bitwidth of read/write data are equal.
- The output of the write data number is configurable. You can choose whether to output the write number data.
- The flag signal output is optional. You can choose to output almost empty flag, almost full flag, or no flag.
- If you choose to output a almost empty or almost full flag, the threshold of almost empty, almost full is configurable. They can be set as static single constant threshold, static double constant threshold, dynamic single input threshold, dynamic double input threshold.
- The ECC check function is optional. The ECC check function can be selected when the asynchronous FIFO internal memory is achieved by

block SRAM and the read/write bitwidth are equal and are below 64 bit.

- The output register function is configurable. If you choose the output register function, read-enable (RdEn) control is optional. If you choose enable-control, the output register is controlled by RdEn, and the last data cannot be output. If you choose the output register function instead of read-enable control, the read data output will be a cycle later, and the last data will be output.
- Supports first-word fall-through function.

2.4 FIFO Maximum Frequency and Resource Utilization

Gowin FIFO is implemented by Verilog. The Max. frequency of FIFO changes according to the data depth, bitwidth, and speed of the devices in use. Performance and resource utilization may vary when you use different devices of varying density, speed, or grades.

Take the GW2A-55 FPGA as an instance. Table 2-2 and Table 2-3 show the FIFO resource utilization and performance for these devices.

Table 2-2 Asynchronous FIFO Without Configuration Options

FIFO Type	Depth x Width	FPGA Family	Performance (MHz)	Resource Utilization			
				CFU Logics	FFs	Block SRAM	Shadow SRAM
Block SRAM	4096 x 16	GW2A-55-484	Wr_clk=120.396；Rd_clk=109.567	251	124	4	0
	512 x 16		Wr_clk=133.227；Rd_clk=116.194	181	96	1	0
	64 x 16		Wr_clk=148.443；Rd_clk=165.963	138	68	1	0
Shadow SRAM	2048 x 16	GW2A-55-484	Wr_clk=98.085；Rd_clk=94.967	5098	126	0	512
	512 x 16		Wr_clk=119.934；Rd_clk=80.903	1361	108	0	128
	64 x 16		Wr_clk=146.482；Rd_clk=176.987	302	88	0	16
LUT	64 x 16	GW2A-55-484	Wr_clk=144.51；Rd_clk=118.609	770	1105	0	0
Block SRAM	4096 x 16	GW2A-18-484	Wr_clk=120.907；Rd_clk=104.532	251	124	4	0

FIFO Type	Depth x Width	FPGA Family	Performance (MHz)	Resource Utilization			
				CFU Logics	FFs	Block SRAM	Shadow SRAM
	512 x 16		Wr_clk=140.547; Rd_clk=116.818	181	96	1	0
	64 x 16		Wr_clk=165.234; Rd_clk=168.747	138	68	1	0
Shadow SRAM	1024x 16	GW2A-18-484	Wr_clk=119.613; Rd_clk=97.107	2588	117	0	256
	512 x 16		Wr_clk=123.976; Rd_clk=92.429	1361	108	0	128
	64 x 16		Wr_clk=155.54; Rd_clk=175.921	302	88	0	16
LUT	64 x 16	GW2A-18-484	Wr_clk=139.892; Rd_clk=117.022	770	1105	0	0
Block SRAM	4096 x 16	GW1N-4-144 GW1N-4-144 GW1N-4-144	Wr_clk=71.261; Rd_clk=64.502	229	127	4	0
	512 x 16		Wr_clk=71.247; Rd_clk=66.017	176	94	1	0
	64 x 16		Wr_clk=81.447; Rd_clk=76.353	144	70	1	0
Shadow SRAM	4096 x 16	GW1N-4-144 GW1N-4-144 GW1N-4-144	Not supported				

FIFO Type	Depth x Width	FPGA Family	Performance (MHz)	Resource Utilization			
				CFU Logics	FFs	Block SRAM	Shadow SRAM
	512 x 16						
	64 x 16						
LUT	64 x 16	GW1N-4-144	Wr_clk=84.709；Rd_clk=57.022	763	1105	0	0

Notes!

- “–” denotes not supported yet. Please wait for subsequent updates.
- Application verification of FIFO on other GOWIN FPGA devices will be released in succession.

Table 2-3 Asynchronous FIFO With Configurtion Options

FIFO Type	Depth x Width	FPGA Family	Performance (MHz)	Resource Utilization				Configuration Item		
				CFU Logics	FFs	Block SRAM	Shadow SRAM	Data Num.	Control Flag	ECC
Block SRAM	4096 x 16	GW2A-55-484	Wr_clk=102.33；Rd_clk=104.918	376	157	6	0	√	√	√
	512 x 16		Wr_clk=140.366；Rd_clk=107.101	302	149	1	0	√	√	√
	64 x 16		Wr_clk=151.163；Rd_clk=148.65	242	112	1	0	√	√	√
Shadow SRAM	2048 x 16	GW2A-55-484	Wr_clk=95.643；Rd_clk=88.557	5106	170	0	512	√	√	×
	512 x 16		Wr_clk=117.17；Rd_clk=84.09	1373	148	0	128	√	√	×

FIFO Type	Depth x Width	FPGA Family	Performance (MHz)	Resource Utilization				Configuration Item		
				CFU Logics	FFs	Block SRAM	Shadow SRAM	Data Num.	Control Flag	ECC
			7							
	64 x 16		Wr_clk=141.822；Rd_clk=143.365	312	116	0	16	√	√	×
LUT	64 x 16	GW2A-55-484	Wr_clk=151.829；Rd_clk=125.791	731	1139	0	0	√	√	×
Block SRAM	4096 x 16	GW2A-18-484	Wr_clk=99.288；Rd_clk=114.648	376	157	6	0	√	√	√
	512 x 16		Wr_clk=122.775；Rd_clk=110.636	302	149	1	0	√	√	√
	64 x 16		Wr_clk=148.711；Rd_clk=173.416	242	112	1	0	√	√	√
Shadow SRAM	1024 x 16	GW2A-18-484	Wr_clk=128.221；Rd_clk=76.014	2600	159	0	256	√	√	×
	512 x 16		Wr_clk=119.455；Rd_clk=87.461	1373	148	0	128	√	√	×
	64 x 16		Wr_clk=146.078；Rd_clk=156.31	312	116	0	16	√	√	×
LUT	64 x 16	GW2A-18-484	Wr_clk=142.922；Rd_clk=106.383	731	1139	0	0	√	√	×
Block SRAM	4096 x 16	GW1N-4-144	Wr_clk=65.951；	385	157	6	0	√	√	√

FIFO Type	Depth x Width	FPGA Family	Performance (MHz)	Resource Utilization				Configuration Item		
				CFU Logics	FFs	Block SRAM	Shadow SRAM	Data Num.	Control Flag	ECC
		GW1N-4-144	Rd_clk=49.374							
	512 x 16	GW1N-4-144	Wr_clk=77.302；Rd_clk=64.869	286	129	1	0	√	√	√
	64 x 16		Wr_clk=75.821；Rd_clk=75.619	266	90	1	0	√	√	√
Shadow SRAM	4096 x 16	GW1N-4-144	Not supported							
	512 x 16	GW1N-4-144								
	64 x 16	GW1N-4-144								
LUT	64 x 16	GW1N-4-144	Wr_clk=84.263；Rd_clk=80.362	717	1139	0	0	√	√	×

Notes!

- “–” denotes not supported yet. Please wait for subsequent updates.
- Application verification of FIFO on other GOWIN FPGA will be released in succession.

2.5 FIFO SC Maximum Frequency and Resource Utilization

Gowin FIFO SC is implemented by Verilog. The Max. frequency of FIFO SC mainly depends on the data depth, bitwidth, and speed of the devices used. Performance and utilization will change when you use different devices of different density, speed, or grades.

Take the GW2A-55 FPGA as an instance. Table 2-4 and Table 2-5 show the FIFO SC resource utilization and performance for these devices.

Table 2-4 FIFO SC Without Configuration Options

FIFO Type	Depth x Width	FPGA Family	Performance (MHz)	Resource Utilization			
				CFU Logics	FFs	Block SRAM	Shadow SRAM
Block SRAM	4096 x 16	GW2A-55-484	131.757	91	42	4	0
	512 x 16		131.803	58	33	1	0
	64 x 16		151.448	65	29	1	0
Shadow SRAM	4096 x 16	GW2A-55-484	82.58	4943	56	0	512
	512 x 16		77.827	1260	50	0	128
	64 x 16		128.136	230	48	0	16
LUT	64 x 16	GW2A-55-484	100.266	639	1064	0	0
Block SRAM	4096 x 16	GW2A-18-484	123.377	91	42	4	0
	512 x 16		135.391	58	33	1	0
	64 x 16		147.623	65	29	1	0
Shadow SRAM	1024 x 16	GW2A-18-484	91.559	2473	53	0	256
	512 x 16		83.219	1260	50	0	128
	64 x 16		135.602	230	48	0	16
LUT	64 x 16	GW2A-18-484	99.671	639	1064	0	0
Block SRAM	4096 x 16	GW1N-4-144 GW1N-4-144 GW1N-4-144	77.268	91	42	4	0
	512 x 16		75.283	57	33	1	0
	64 x 16		70.792	73	30	1	0
Shadow SRAM	4096 x 16	GW1N-4-144 GW1N-4-144	Not Supported				
	512 x 16						
	64 x 16						
LUT	64 x 16	GW1N-4-144	56.535	640	1065	0	0

Notes!

- “–” denotes not supported yet. Please wait for subsequent updates.
- Application verification of FIFO SC on other GOWIN FPGA devices will be released in succession.

Table 2-5 FIFO SC With Configuration Options

FIFO Type	Depth x Width	FPGA Family	Performance (MHz)	Resource Utilization				Configuration Item		
				CFU Logics	FFs	Block SRAM	Shadow SRAM	Data Num.	Control Flag	ECC
Block SRAM	4096 x 16	GW2A-55-484	121.158	187	77	6	0	√	√	√
	512 x 16		133.801	154	67	1	0	√	√	√
	64 x 16		147.675	168	60	1	0	√	√	√
Shadow SRAM	2048 x 16	GW2A-55-484	82.237	4943	84	0	512	√	√	×
	512 x 16		91.368	1260	76	0	128	√	√	×
	64 x 16		125.148	230	71	0	16	√	√	×
LUT	64 x 16	GW2A-55-484	101.459	639	1087	0	0	√	√	×
Block SRAM	4096 x 16	GW2A-18-484	123.42 132.574 150.944	187	77	6	0	√	√	√
	512 x 16			154	67	1	0	√	√	√
	64 x 16			168	60	1	0	√	√	√
Shadow SRAM	1024 x 16	GW2A-18-484	96.128 80.138 137.403	2473	80	0	256	√	√	×
	512 x 16			1260	76	0	128	√	√	×
	64 x 16			230	71	0	16	√	√	×
LUT	64 x 16	GW2A-18-484	106.956	639	1087	0	0	√	√	×
Block SRAM	4096 x 16	GW1N-4-144	71.363	201	69	6	0	√	√	√
	512 x 16		65.051	168	67	1	0	√	√	√

FIFO Type	Depth x Width	FPGA Family	Performance (MHz)	Resource Utilization				Configuration Item		
				CFU Logics	FFs	Block SRAM	Shadow SRAM	Data Num.	Control Flag	ECC
	64 x 16		81.128	185	61	1	0	√	√	√
Shadow SRAM	4096 x 16	GW1N-4-144	Not supported							
	512 x 16									
	64 x 16									
LUT	64 x 16	GW1N-4-144	67.522	640	1088	0	0	√	√	×

Note!

- “–” denotes not supported yet. Please wait for subsequent updates.
- Application verification of FIFO SC on other GOWIN FPGA devices will be released in succession.

3 Port Description

3.1 Gowin FIFO IP I/O Port

The descriptions of Gowin FIFO I/O IP are listed in Table 3-1.

Table 3-1 Gowin FIFO IP I/O List

Signal	Data Width	I/O	Initial State	Optional	Description
Data	[WDSIZE-1:0]	I	-	No	Input data
WrClk	1	I	-	No	Write clock
RdClk	1	I	-	No	Read clock
WrEn	1	I	-	No	Write enable
RdEn	1	I	-	No	Read enable
Reset	1	I	-	Yes	Reset, active high
WrReset	1	I	-	Yes	Write reset, active high
RdReset	1	I	-	Yes	Read reset, active high
AlmostEmptySetTh	[RASIZE-1:0]	I	-	Yes	Almost empty flag, set threshold to 1
AlmostEmptyClrTh	[RASIZE-1:0]	I	-	Yes	Almost empty flag, set threshold to 0
AlmostEmptyTh	[RASIZE-1:0]	I	-	Yes	Almost empty flag, set threshold to 1
AlmostFullSetTh	[ASIZE-1:0]	I	-	Yes	Almost full flag, set threshold to 1
AlmostFullClrTh	[ASIZE-1:0]	I	-	Yes	Almost full flag, set threshold to 0
AlmostFullTh	[ASIZE-1:0]	I	-	Yes	Almost full flag, set threshold to 1
Q	[RDSIZE-1:0]	O	-	No	Read data

Signal	Data Width	I/O	Initial State	Optional	Description
Empty	1	O	1	No	Empty flag
Full	1	O	0	No	Full flag
Wnum	[ASIZE: 0]	O	0	Yes	Write data number
Rnum	[RASIZE: 0]	O	0	Yes	Readable data number
Almost_Empty	1	O	1	Yes	Almost empty flag
Almost_Full	1	O	0	Yes	Almost full flag
ERROR	2	O	0	Yes	ECC check output

3.2 Gowin FIFO SC IP I/O Port

The descriptions of Gowin FIFO SC IP I/O are listed in Table 3-2.

Table 3-2 Gowin FIFO SC IP I/O List

Signal	Data Width	I/O	Initial State	Optional	Description
Data	[DSIZE-1:0]	I	-	No	Input data
Clk	1	I	-	No	Write clock
WrEn	1	I	-	No	Write enable
RdEn	1	I	-	No	Read enable
Reset	1	I	-	Yes	Reset, active high.
AlmostEmptySetTh	[ASIZE-1:0]	I	-	Yes	Almost empty flag, set threshold to 1.
AlmostEmptyClrTh	[ASIZE-1:0]	I	-	Yes	Almost empty flag, set threshold to 0.
AlmostEmptyTh	[ASIZE-1:0]	I	-	Yes	Almost empty flag, set threshold to 1.
AlmostFullSetTh	[ASIZE-1:0]	I	-	Yes	Almost full flag, set threshold to 1
AlmostFullClrTh	[ASIZE-1:0]	I	-	Yes	Almost full flag, set threshold to 0.
AlmostFullTh	[ASIZE-1:0]	I	-	Yes	Half-full flag, set threshold to 1.
Q	[DSIZE-1:0]	O	-	No	Read data
Empty	1	O	1	No	Empty flag
Full	1	O	0	No	Full flag
Wnum	[ASIZE: 0;	O	0	Yes	Write data number
Almost_Empty	1	O	1	Yes	Almost empty flag
Almost_Full	1	O	0	Yes	Almost full flag

Signal	Data Width	I/O	Initial State	Optional	Description
ERROR	2	O	0	Yes	ECC check output

4 Timing Description

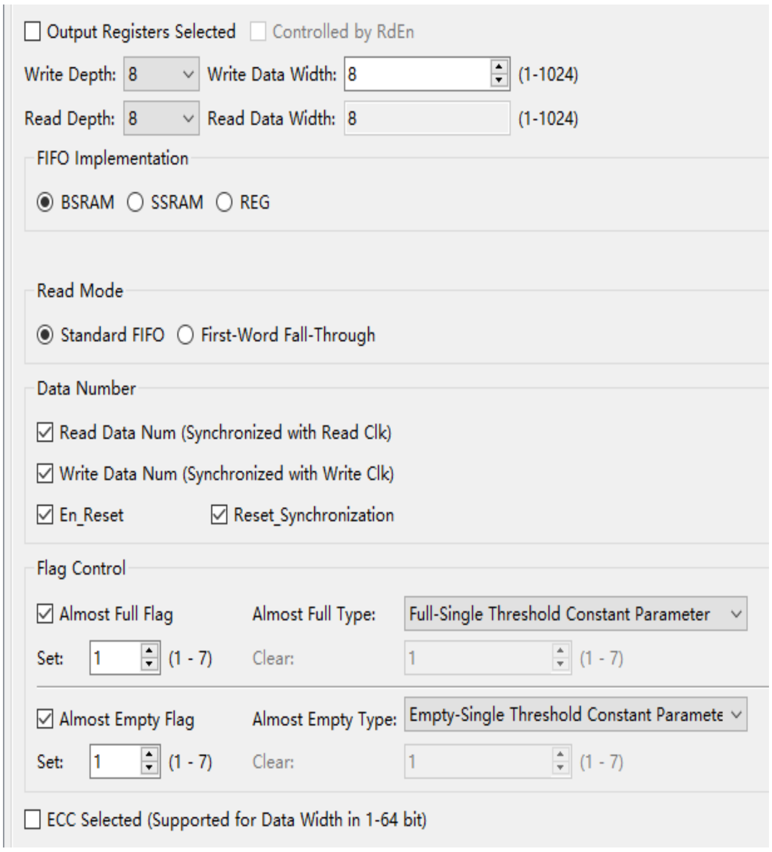
This chapter describes the FIFO and FIFO SC timing during read and write operations.

In practical applications, you may configure different output control signals according to your requirements. The Read/Write control occasions are different. The most common and important signals including input, output, empty & full, almost empty, almost full are described in this chapter.

4.1 FIFO Signal Timing

Figure 4-2 shows a read/write operation of the FIFO with the configurations shown in Figure 4-1.

Figure 4-1 FIFO Configuration



As shown in Figure 4-2, when the FIFO is not full, the write enable is pulled up and the data is written to FIFO. The FIFO is full after eight writes and the write enable will be masked. The Full signal is pulled up, and data cannot be written at this time. When the FIFO is not empty, the read enable is pulled up and the data written to the FIFO is read to Q in turn. When the eighth number is read, the read enable will be masked. The Empty signal is pulled up, and the data cannot be read at this time.

Figure 4-2 FIFO IP Timing

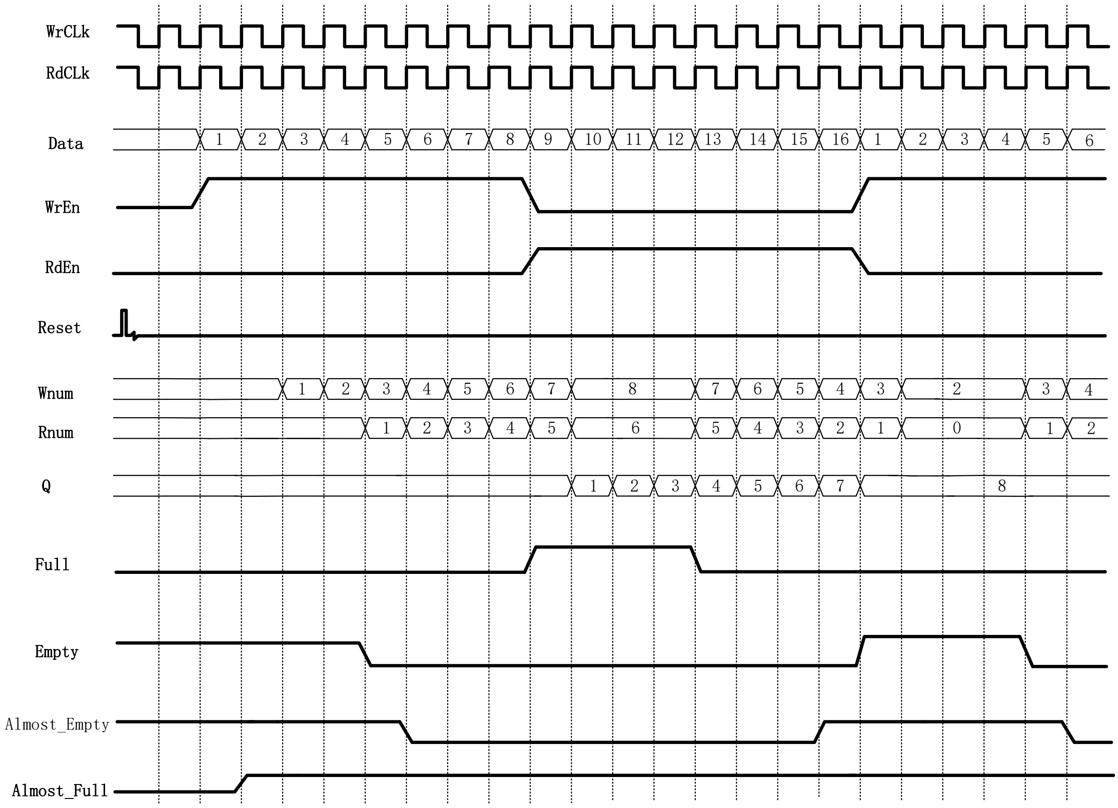
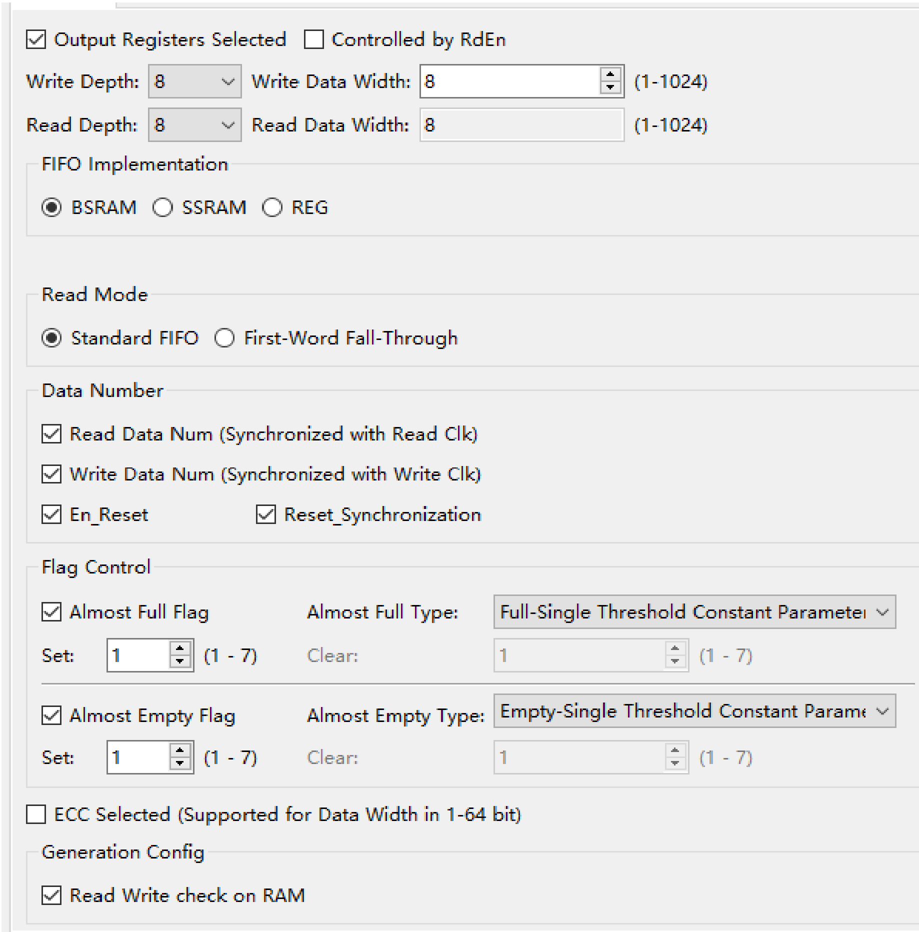


Figure 4-4 shows a read/write operation of the FIFO with the configurations shown in Figure 4-3.

Figure 4-3 FIFO Configuration



As shown in Figure 4-3, the output register is selected, so the read data is one cycle later than when the output register is not configured, and the last data will be output.

Figure 4-4 FIFO IP Timing

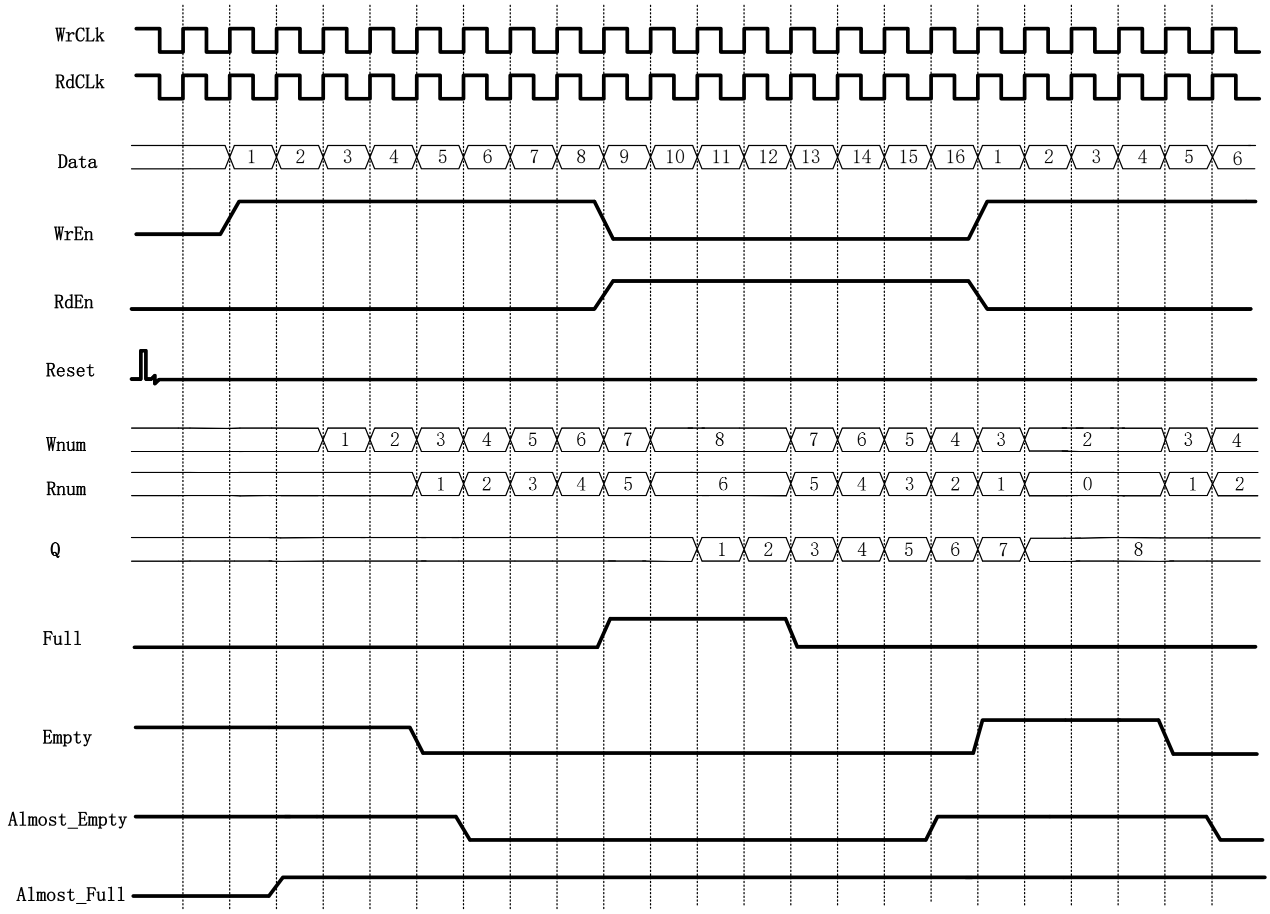
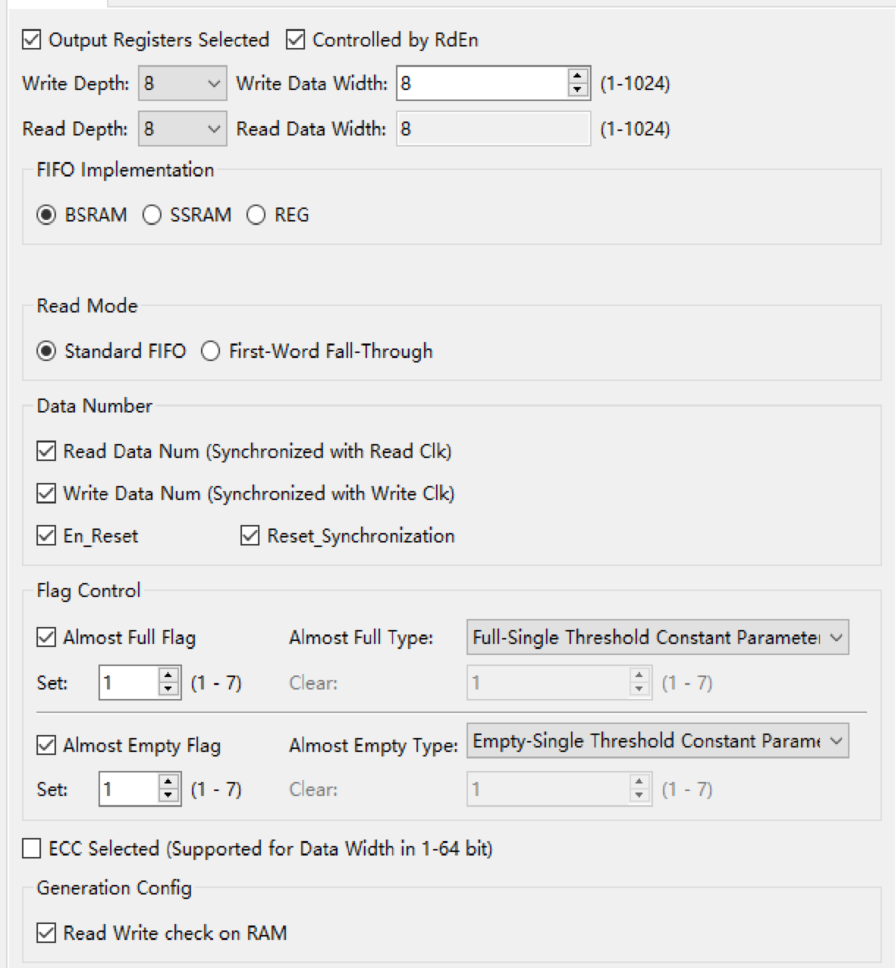


Figure 4-6 shows a read/write operation of the FIFO with the configurations shown in Figure 4-5.

Figure 4-5 FIFO Configuration



As shown in Figure 4-5, the output register is selected and the read enable control is also selected, i.e., the output register is controlled by RdEn, so the read data is one cycle later than when the output register is not configured, and the last data will not be output under the first read enable, and this data will be the first data output of the next read enable.

Figure 4-6 FIFO IP Timing

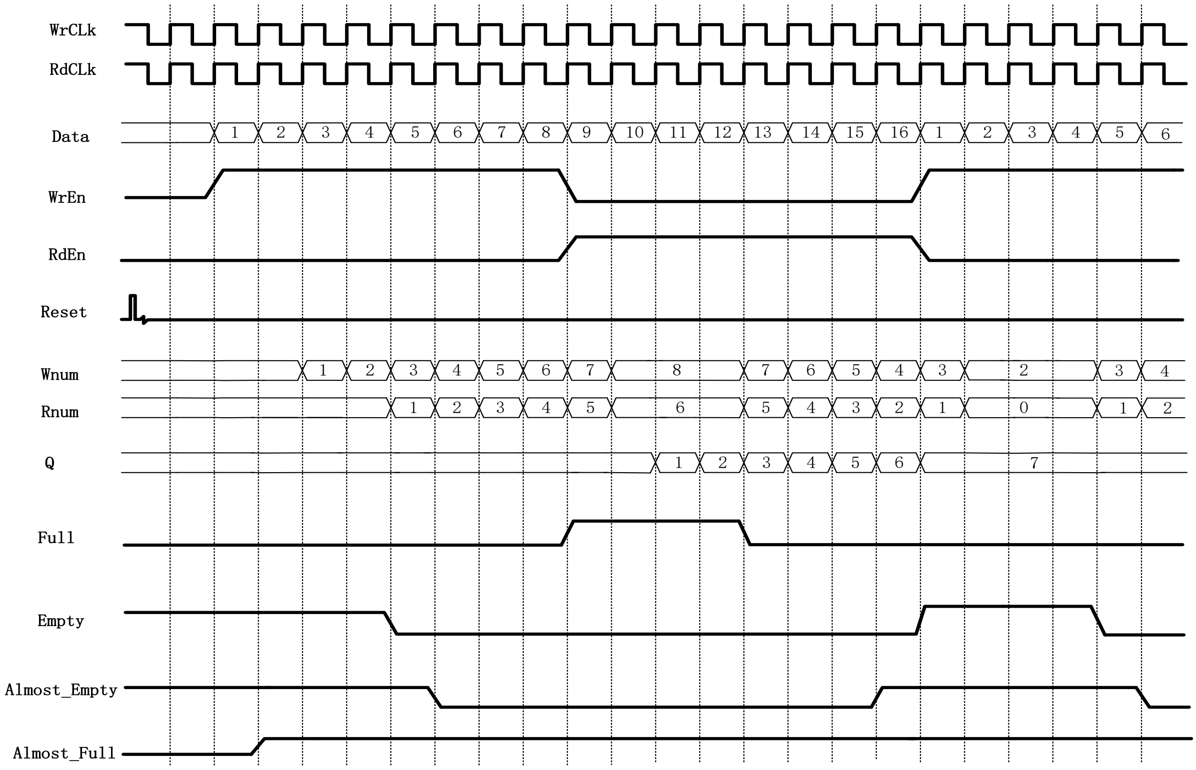
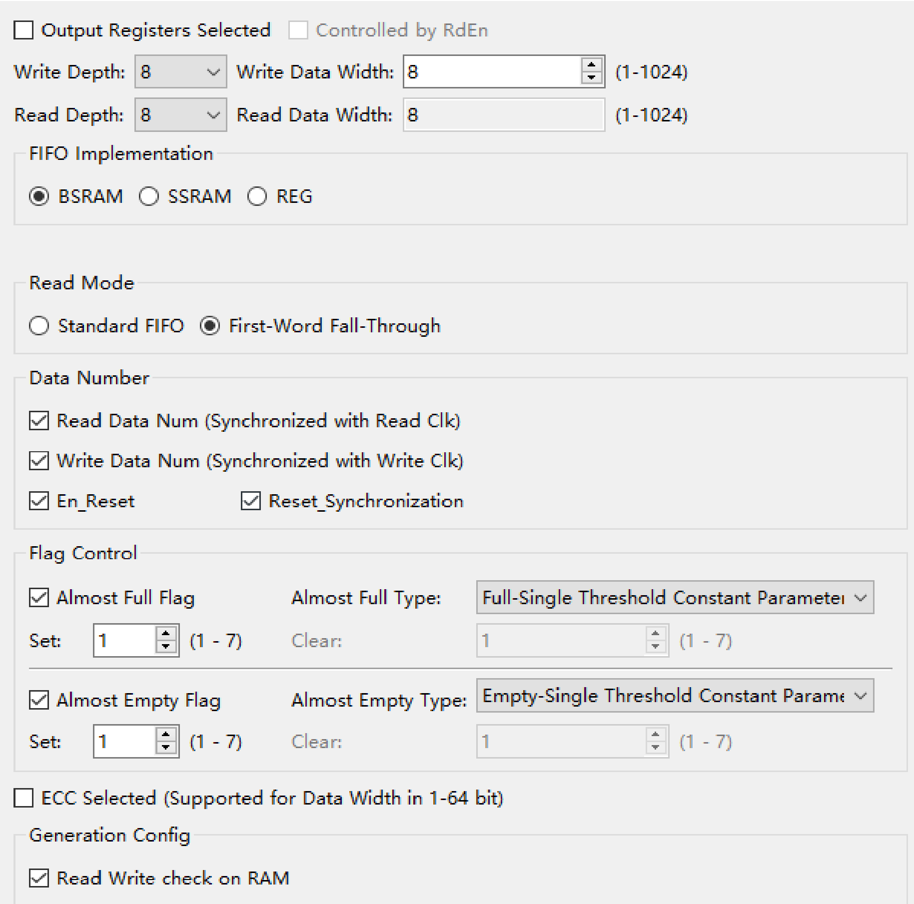


Figure 4-8 shows a read/write operation of the FIFO with the configurations shown in Figure 4-7.

Figure 4-7 FIFO Configuration



As shown in Figure 4-7, First-Word Fall-Through selected, it will put the first number written on the output data bus immediately when the FIFO is non-empty regardless of whether there is a read enable signal or not, and it will output the other data written in order when the read enable is pulled up.

Figure 4-8 FIFO IP Timing

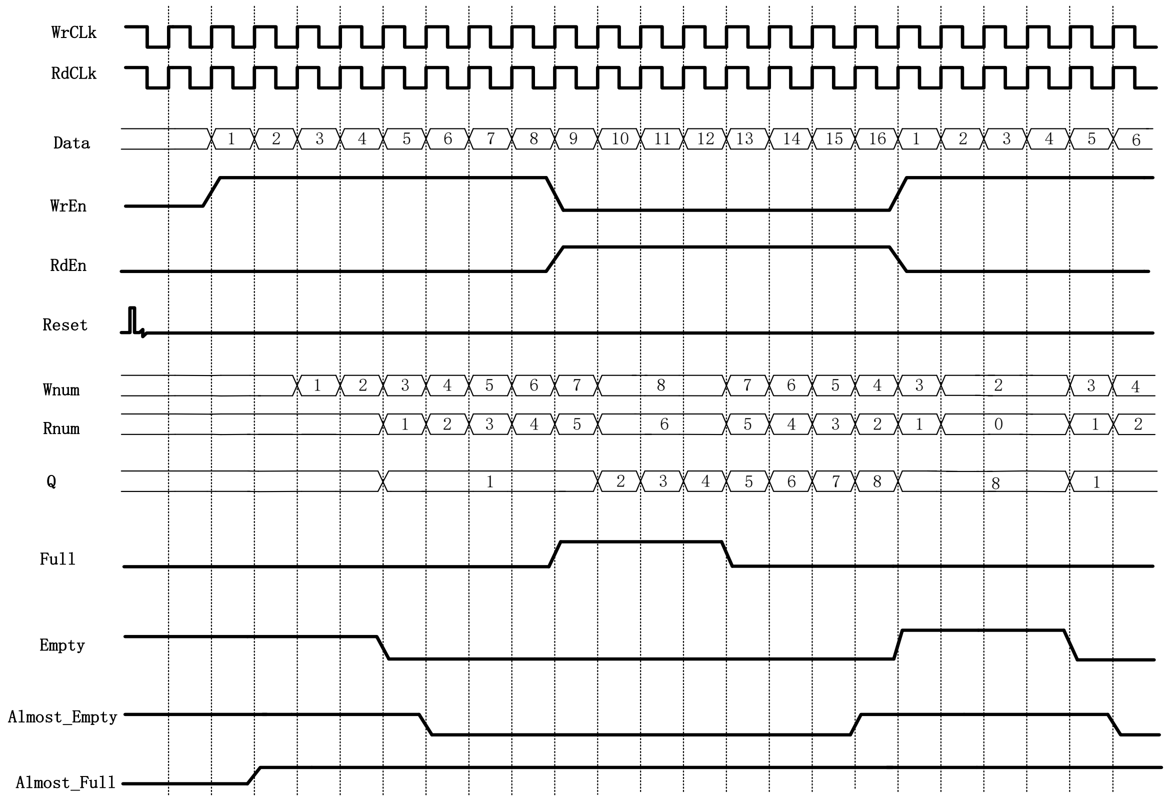
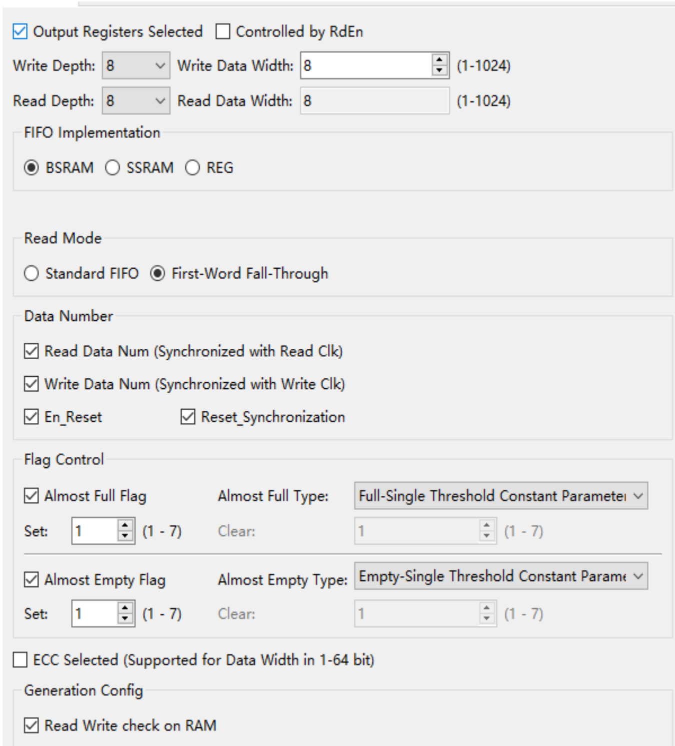


Figure 4-9 shows a read/write operation of the FIFO with the configurations shown in Figure 4-8.

Figure 4-9 FIFO Configuration



As in Figure 4-8, the FWFT and output register are selected, which means that the output data of FIFO in FWFT mode is registered at one level before output.

Figure 4-10 FIFO IP Timing

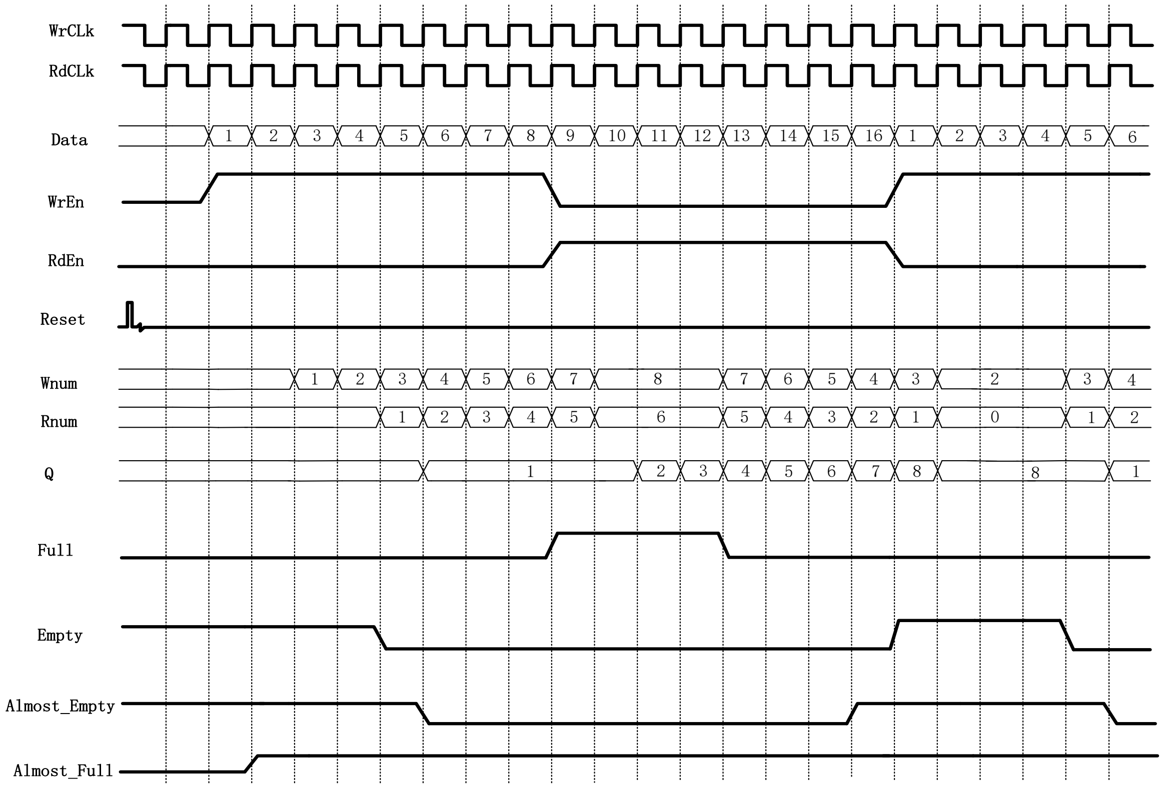
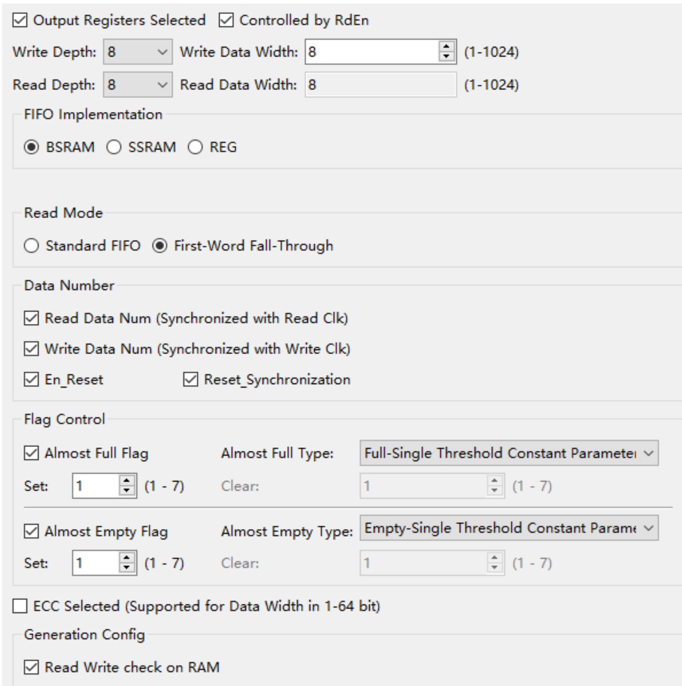


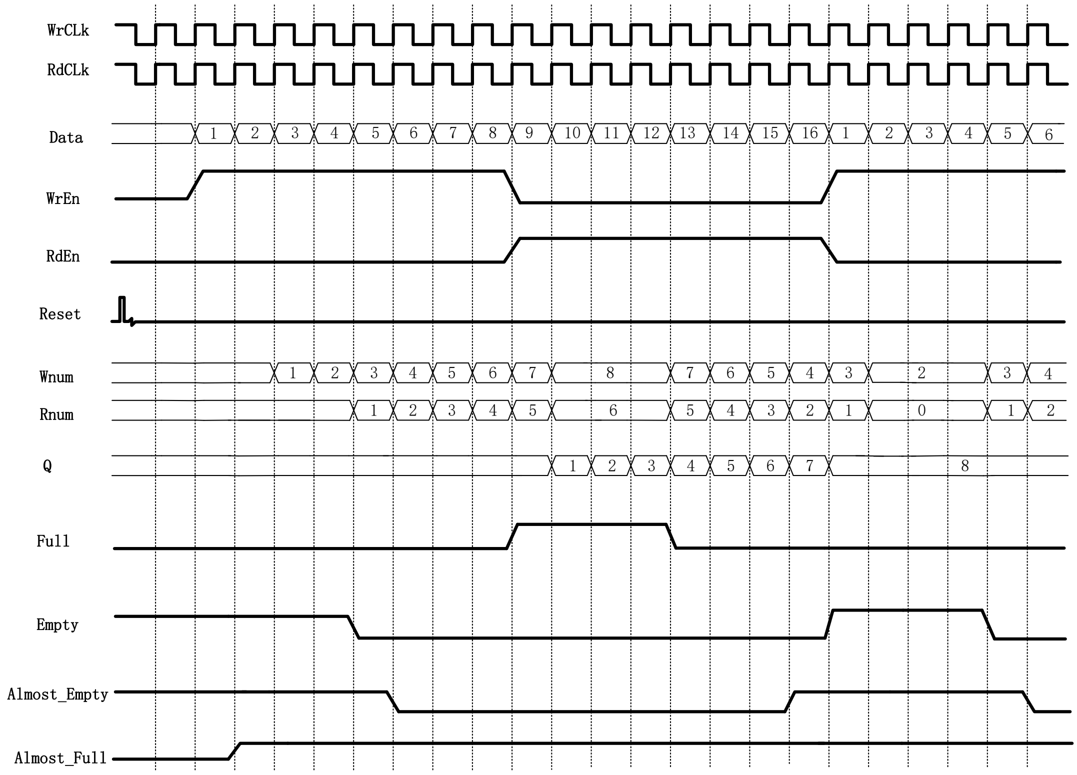
Figure 4-12 shows a read/write operation of the FIFO in the configuration shown in Figure 4-11.

Figure 4-11 FIFO Configuration



As shown in Figure 4-11, the FWFT mode, the output register and read enable control are selected, i.e., the output data of FIFO in FWFT mode is registered at one level, but the registered data is not output immediately, but output under the control of read enable.

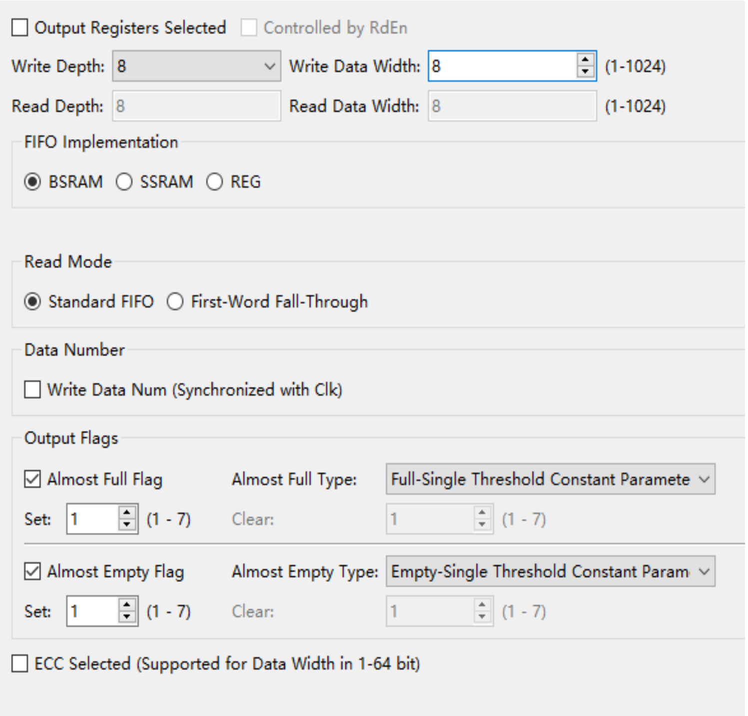
Figure 4-12 FIFO IP Timing



4.2 FIFO SC Signal Timing

Figure 4-14 shows a read/write operation of the FIFO SC with the configurations shown in Figure 4-13.

Figure 4-13 FIFO SC Configuration



As shown in Figure 4-13, when the FIFO is not full, the write enable is pulled up and the data is written to the FIFO; when the FIFO is full after eight writes, the write enable will be masked and the Full signal is pulled up, and no more data can be written at this time. The Empty signal is pulled up, and no more data can be read at this time.

Figure 4-14 FIFO SC IP Timing

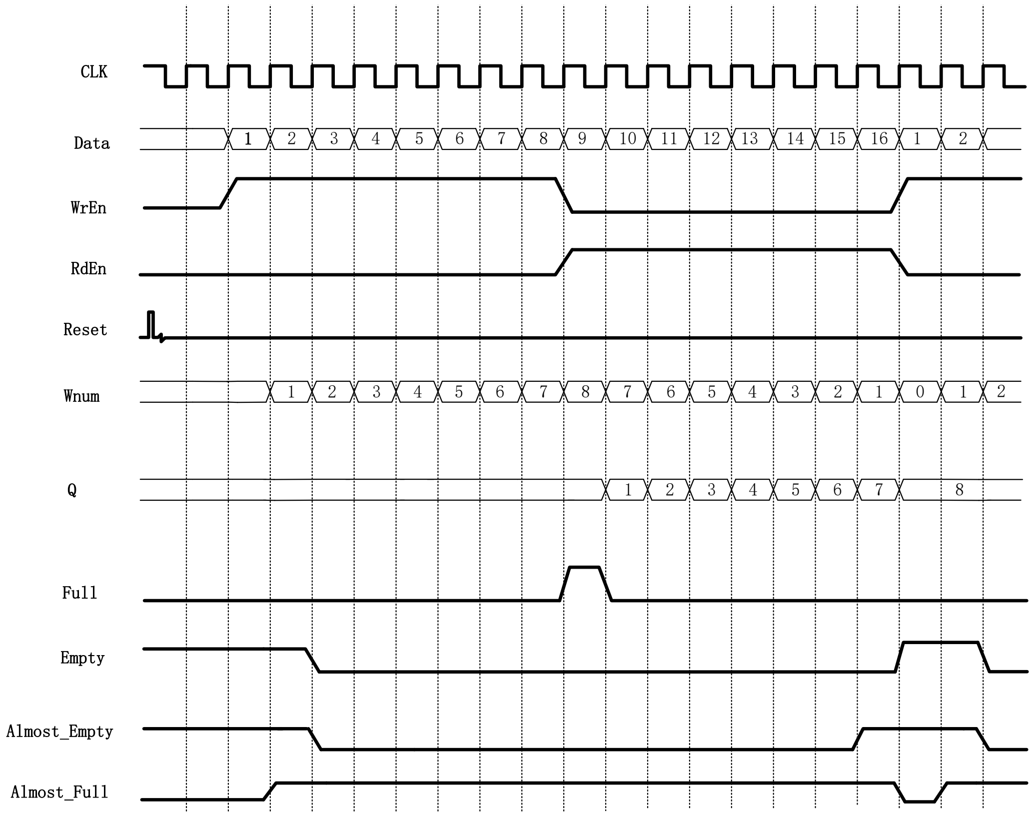
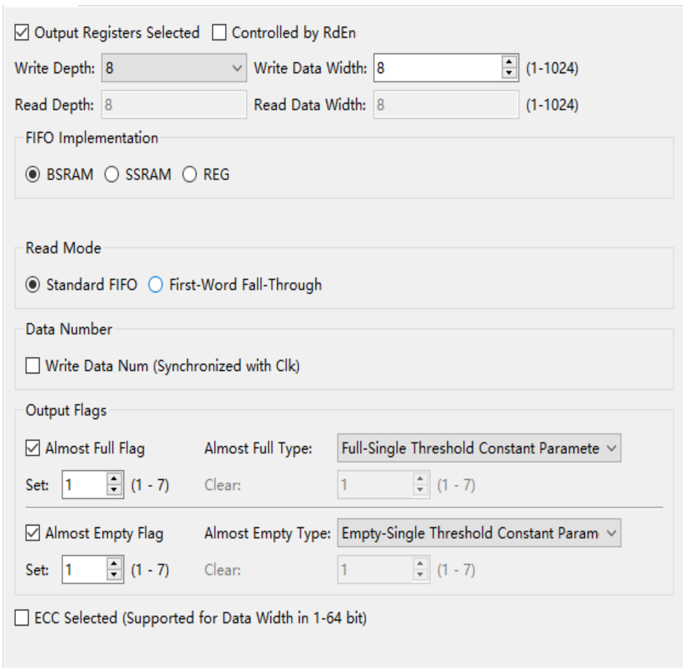


Figure 4-16 shows a read/write operation of the FIFO SC with the configurations shown in Figure 4-15.

Figure 4-15 FIFO SC Configuration



As shown in Figure 4-15, the output register is selected, so the read data is one cycle later than when the output register is not configured, and the last data will be output.

Figure 4-16 FIFO SC IP Timing

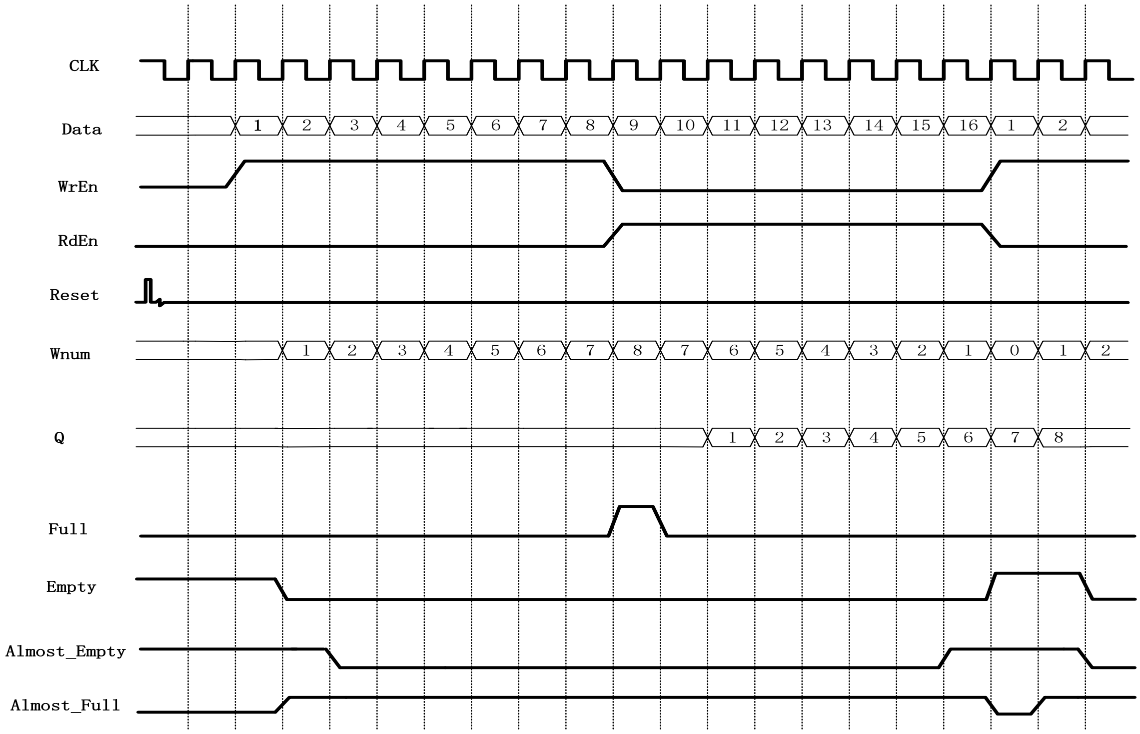
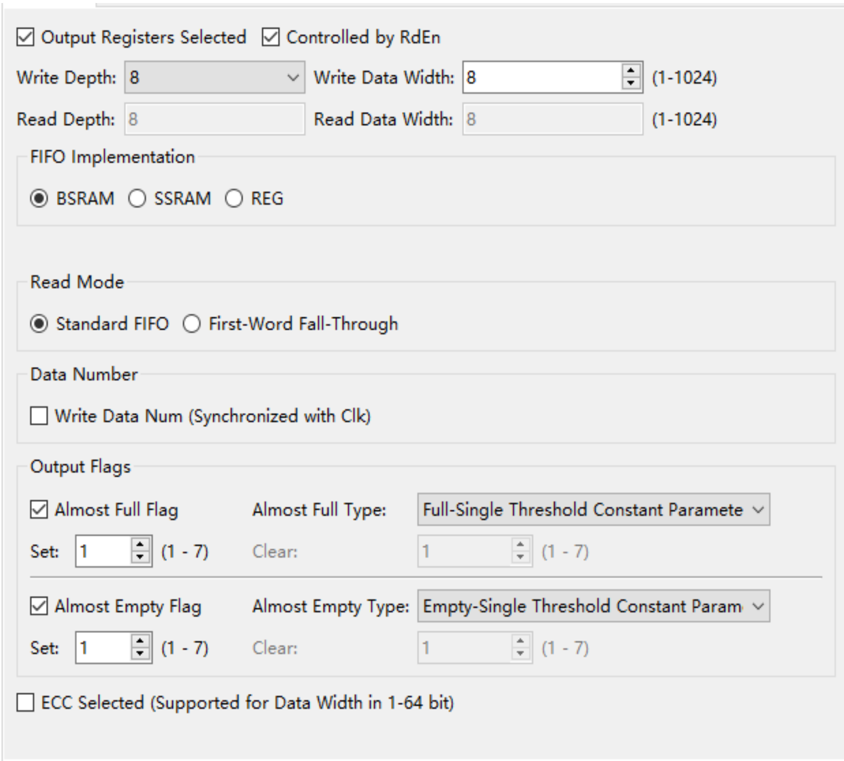


Figure 4-18 shows a read/write operation of the FIFO SC with the configurations shown in Figure 4-17.

Figure 4-17 FIFO SC Configuration



As shown in Figure 4-17, the output register and the read enable control are selected, i.e., the output register is controlled by Raden, so the read data is one cycle later than when the output register is not configured, and the last data will not be output under the first read enable, and this data will be the first data output of the next read enable

Figure 4-18 FIFO SC IP Timing

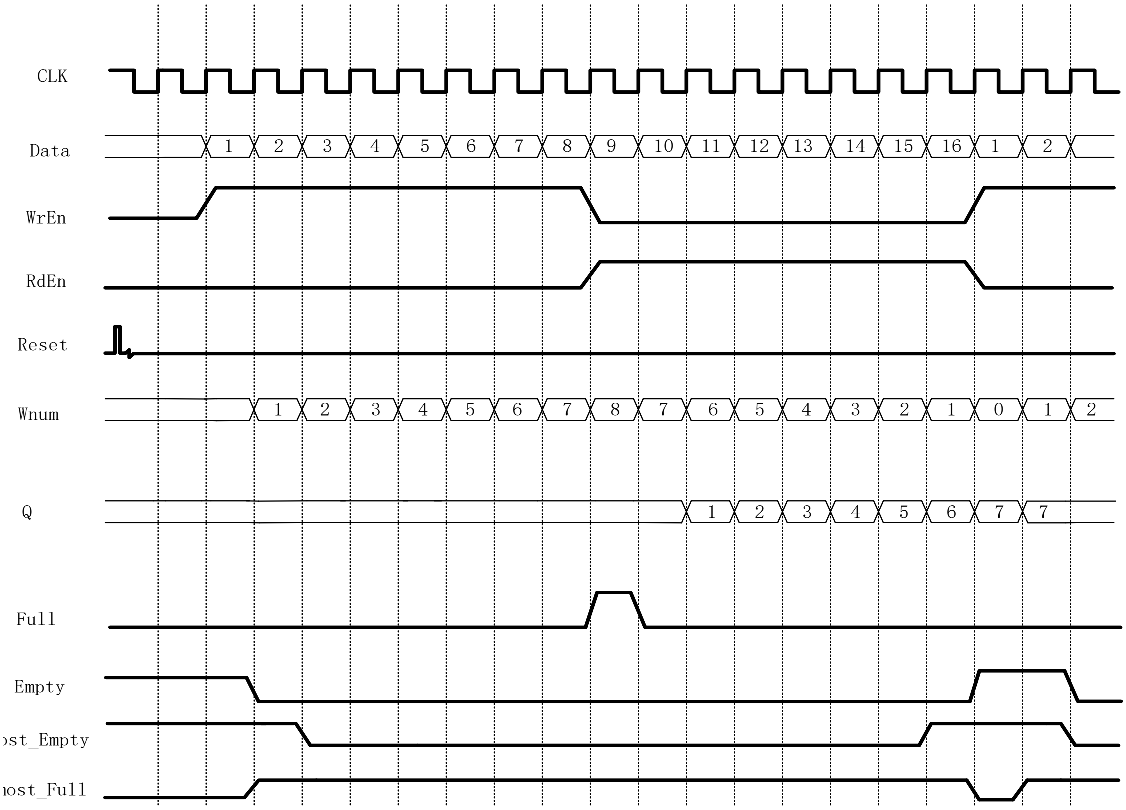
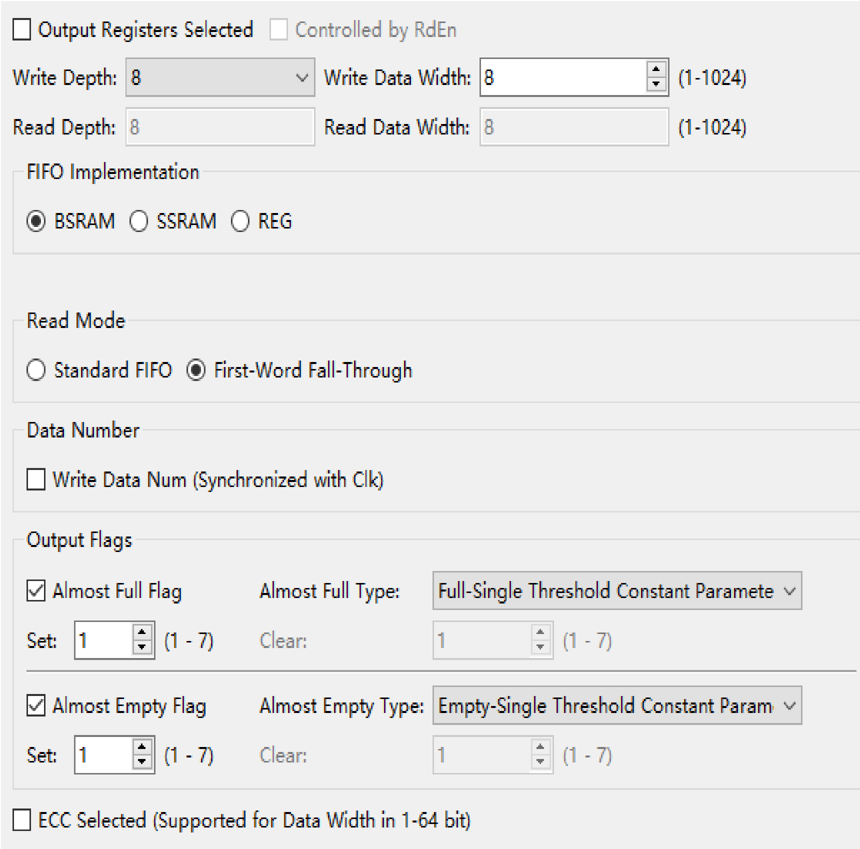


Figure 4-20 shows a read/write operation of the FIFO SC with the configurations shown in Figure 4-19.

Figure 4-19 FIFO SC Configuration



As shown in Figure 4-19, select the FWFT mode. When the FIFO is non-empty, the first data written will be immediately put on the output data bus regardless of whether there is a the read enable signal or not, and the rest of the data written will be output in order when the read enable is pulled up.

Figure 4-20 FIFO SC IP Timing

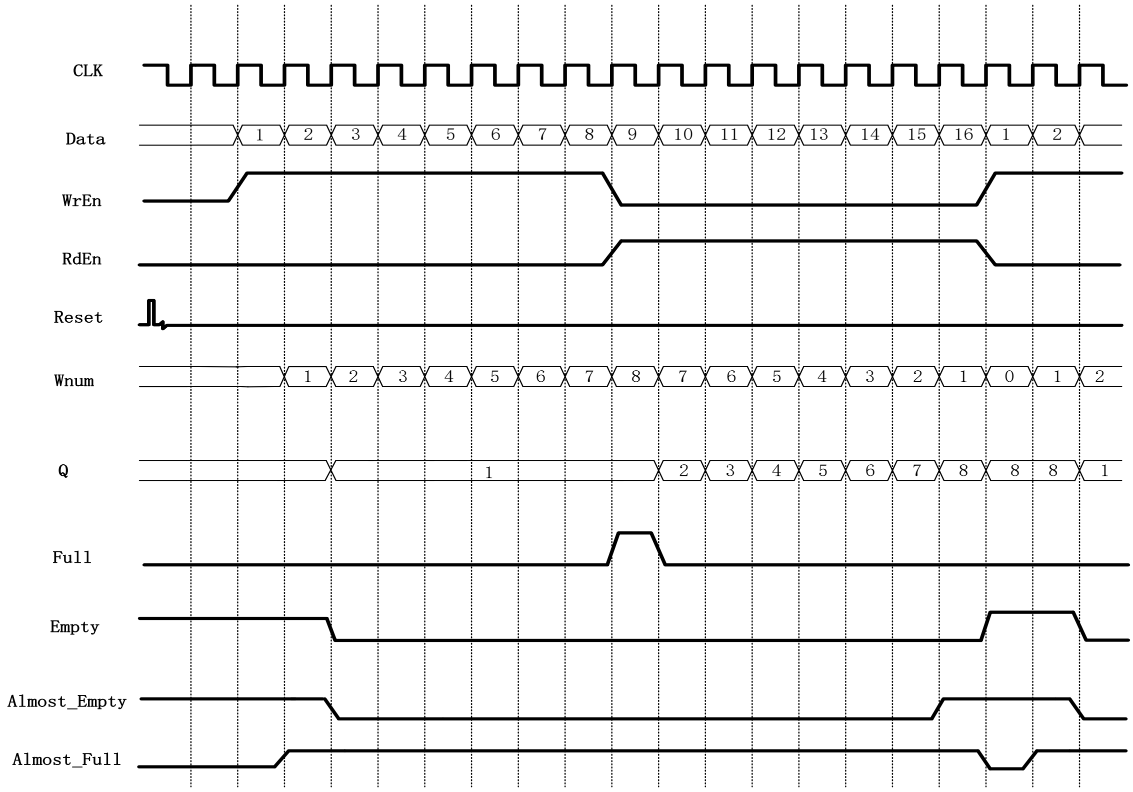
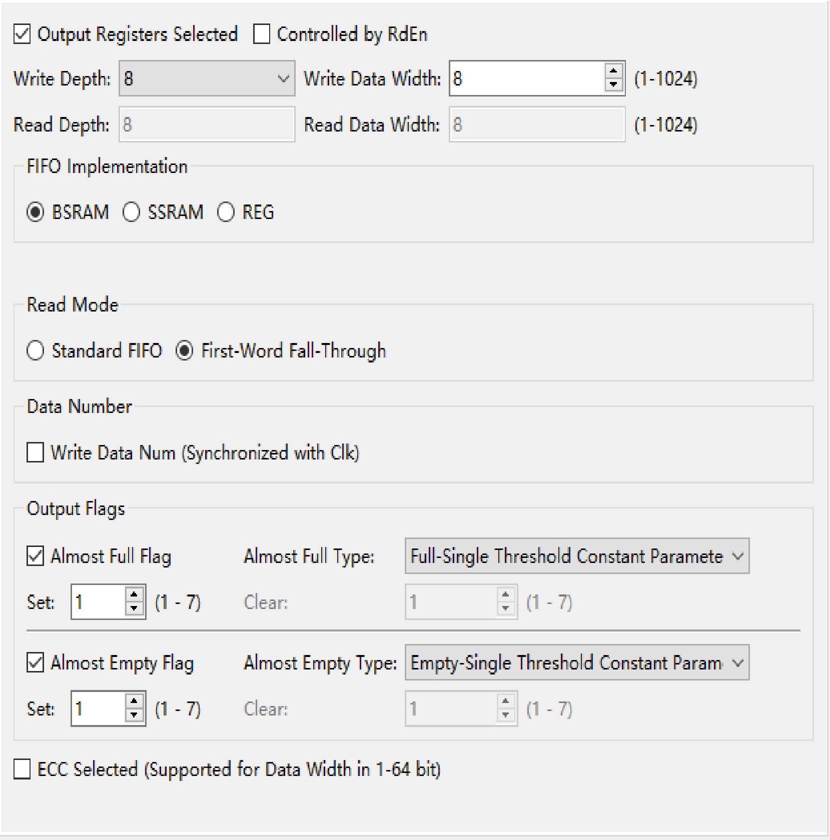


Figure 4-22 shows a read/write operation of the FIFO SC with the configurations shown in Figure 4-21.

Figure 4-21 FIFO SC Configuration



As shown in the Figure 4-21, the FWFT mode and output register are selected, i.e., the output data of FIFO in FWFT mode is registered at one level before output.

Figure 4-22 FIFO SC IP Timing

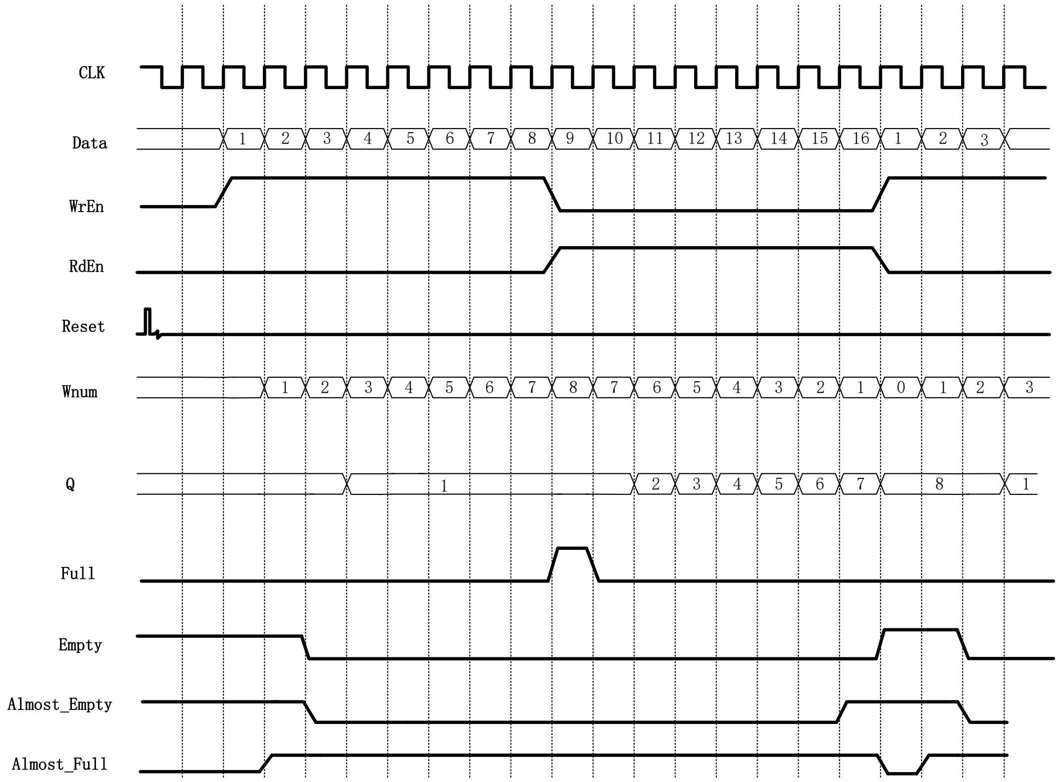
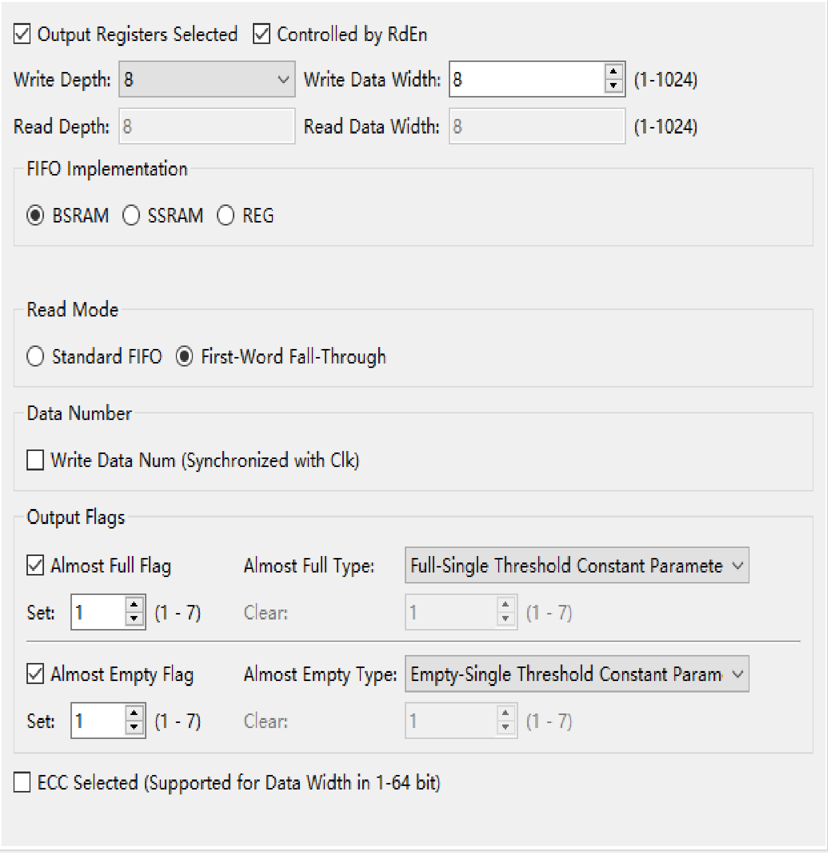


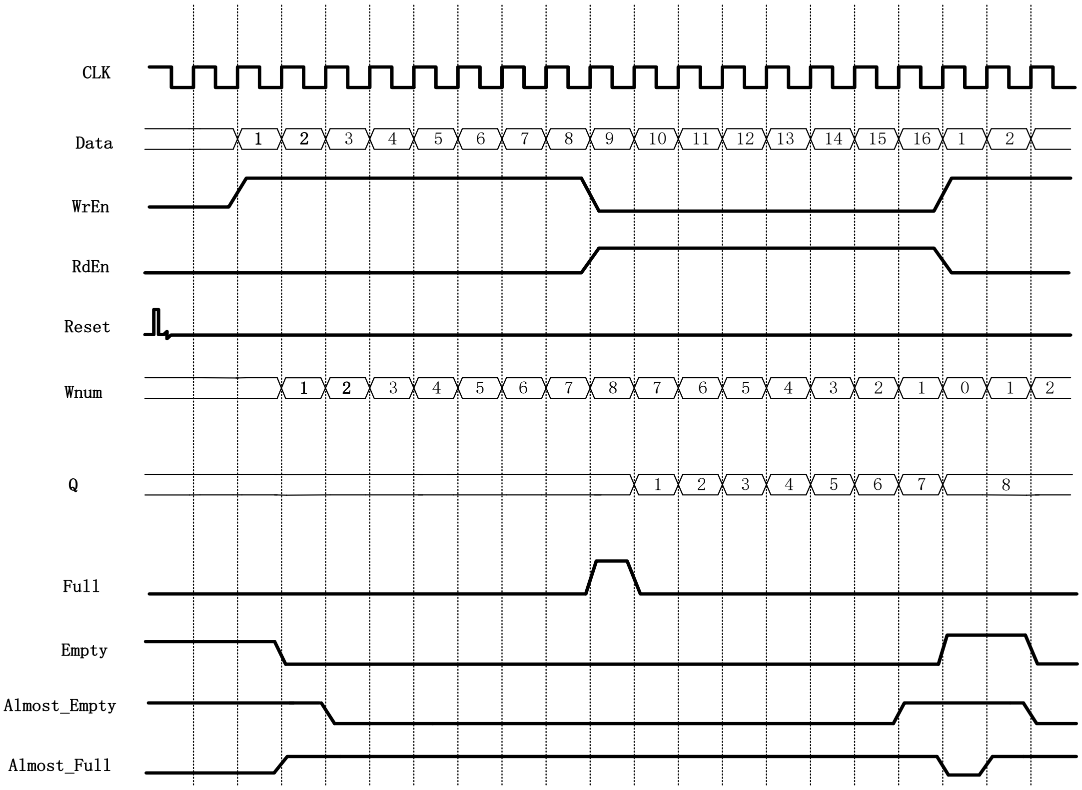
Figure 4-24 shows a read/write operation of the FIFO SC with the configurations shown in Figure 4-23.

Figure 4-23 FIFO SC Configuration



As shown in Figure 4-23, the FWFT mode, output register and read enable control are selected, that is, the output data of FIFO in FWFT mode is registered at one level. The registered data is not output immediately, but output under the control of read enable.

Figure 4-24 FIFO SC IP Timing



5 Gowin FIFO/FIFO SC IP Configuration

Start "IP Core Generator" from the "Tools" menu of Gowin Software, then configure and generate FIFO IP/FIFO SC IP. For the descriptions of FIFO/FIFO SC configuration and parameters, please refer to 5.1 Gowin FIFO IP Configuration and 5.2 FIFO SC IP Configuration.

5.1 Gowin FIFO IP Configuration

Figure 5-1 shows Gowin FIFO IP configuration window.

Figure 5-1 FIFO IP Configuration

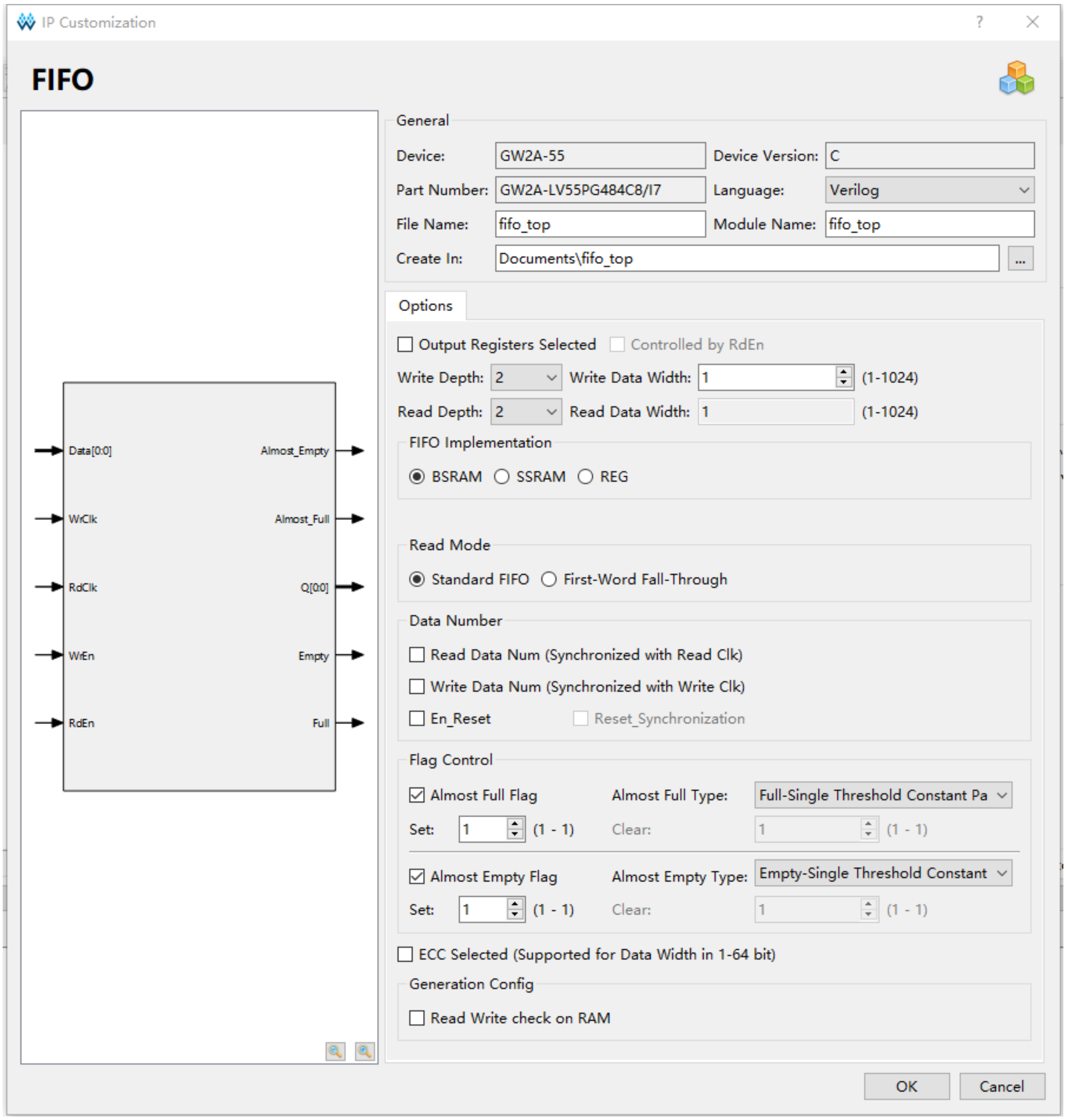


Table 5-1 Gowin FIFO IP Parameters

Tab	Name	Description	Remarks
General	Device	Select a device	-
	Device Version	Select a version	-
	Part Number	Select a Part Number	-
	Language	Select a language	-
	File Name	FIFO files name	-
	Module Name	FIFO top module name	-
	Create In	-	When a FIFO IP model

Tab	Name		Description	Remarks
				is created, you can choose whether to add it to the current project.
Options	FIFO Implementation	BSRAM	Select to use block SRAM, Shadow SRAM, or REG	-
		SSRAM		
		REG		
	Write Depth		Write data depth	When asynchronous FIFO IP write data bitwidth is different from the read data bitwidth, read data bitwidth will be calculated automatically using the following formula: Write Depth x Write Data Width / Read Depth.
	Write Data Width		Write data width	
	Read Depth		Read data depth	
	Read Data Width		Read data width	
	Read Mode	Standard FIFO	Standard FIFO	The standard FIFO reads and writes timing according to the standard timing diagram. The FWFT FIFO will write the first data on the output data bus immediately, regardless of whether there is a read enable signal. After the read enable is pulled high, the additional data will be outputted in order.
		First-Word Fall-Through	FWFT FIFO	
	Data Num	Write Data Num	Write data number	Add Wnum output when it is valid.
		Read Data Num	Read data number	Add Rnum output when it is valid.
		En_Reset	Enable reset	When enable, read/write reset respectively, add WrReset and RdReset input.
		Reset_Synchronization	Same reset	When En_Reset is

<table>
<tr><th>Tab</th><th colspan="3">Name</th><th>Description</th><th>Remarks</th></tr>
<tr><td rowspan="11"></td><td></td><td colspan="2"></td><td></td><td>valid, Reset_Synchronization is optional; If selected, it indicates that a reset is used to add the input Reset.</td></tr>
<tr><td rowspan="10">Flag Control</td><td colspan="2">Almost Full Flag</td><td>Half-full flag enable</td><td>Add Almost_Full output when it is valid.</td></tr>
<tr><td rowspan="4">Almost Full Type</td><td>Full-Single Threshold Constant Parameters</td><td>When static half-full single threshold is valid, Set is valid.</td><td rowspan="4">When almost full flag is valid, users can select one of four options: Full-Single Threshold Constant Parameter, Full-Dual Threshold Constant Parameters, Full-Single Threshold Input Parameter, and Full-Dual Threshold Input Parameters.</td></tr>
<tr><td>Full-Dual Threshold Constant Parameters</td><td>When static half-full dual threshold is valid, Set and Clear are valid.</td></tr>
<tr><td>Full-Single Threshold Input Parameters</td><td>When Dynamic half-full single threshold input is valid, add AlmostFullTh input.</td></tr>
<tr><td>Full-Dual Threshold Input Parameters</td><td>When Dynamic half-full dual threshold input is valid, add AlmostFullSetTh and AlmostFullClrTh input.</td></tr>
<tr><td colspan="2">Set</td><td>Set threshold to 1 when half full</td><td rowspan="2">● Select single constant threshold, Set is valid
● Select dual constant threshold, Set and Clear are valid.</td></tr>
<tr><td colspan="2">Clear</td><td>Set threshold to 0 when half full</td></tr>
<tr><td colspan="2">Almost Empty Flag</td><td>Half-empty flag enable</td><td>Add Almost_Empty output when it is valid.</td></tr>
<tr><td rowspan="2">Almost Empty Type</td><td>Empty-Single Threshold Constant Parameter</td><td>When static half-empty single threshold is valid, Set is valid.</td><td rowspan="2">-</td></tr>
<tr><td>Empty-Dual Threshold Constant</td><td>When static half-empty dual threshold is valid, Set and Clear are valid.</td></tr>
</table>

Tab	Name			Description	Remarks
			Parameters		
			Empty-Single Threshold Input Parameter	When Dynamic half-empty single threshold input is valid, add AlmostEmptyTh input.	
			Empty-Dual Threshold Input Parameter	When dynamic half-empty dual threshold input is valid, add AlmostEmptySetTh and AlmostEmptyClrTh input.	
		Set		Set threshold to 1 when half empty	● Select single constant threshold, Set is valid ● Select dual constant threshold, Set and Clear are valid.
		Clear		Set threshold to 0 when half empty	
	ECC Selected			ECC	Valid when data width is 1-64 bit. Add ERROR output.
	Output Register Selected			Output register can be selected.	When it is valid, data output will be a cycle later. BSRAM output needs to meet the delay from clock to output. When Output Register enable is on, the delay will be met automatically; when Output Register enable is off, the delay from clock to output needs to be met to avoid sampling metastable data.
	Controlled by RdEn			Output is controlled by RdEn.	When Output Register Selected is valid, it can be selected.

Notes!

- To enable the ECC function, data bitwidth needs to be less than or equal to 64-bit.
- Different output control signals and thresholds can be configured according to the requirements. Table 5-1 shows the FIFO IP configuration.
- The "Read Write check on RAM" function is to prevent read/write conflicts on RAM. Please refer to the RAM related manuals or GowinSynthesis manual.

5.2 FIFO SC IP Configuration

Figure 5-2 shows Gowin FIFO SC IP configuration window.

Figure 5-2 FIFO SC IP Configuration

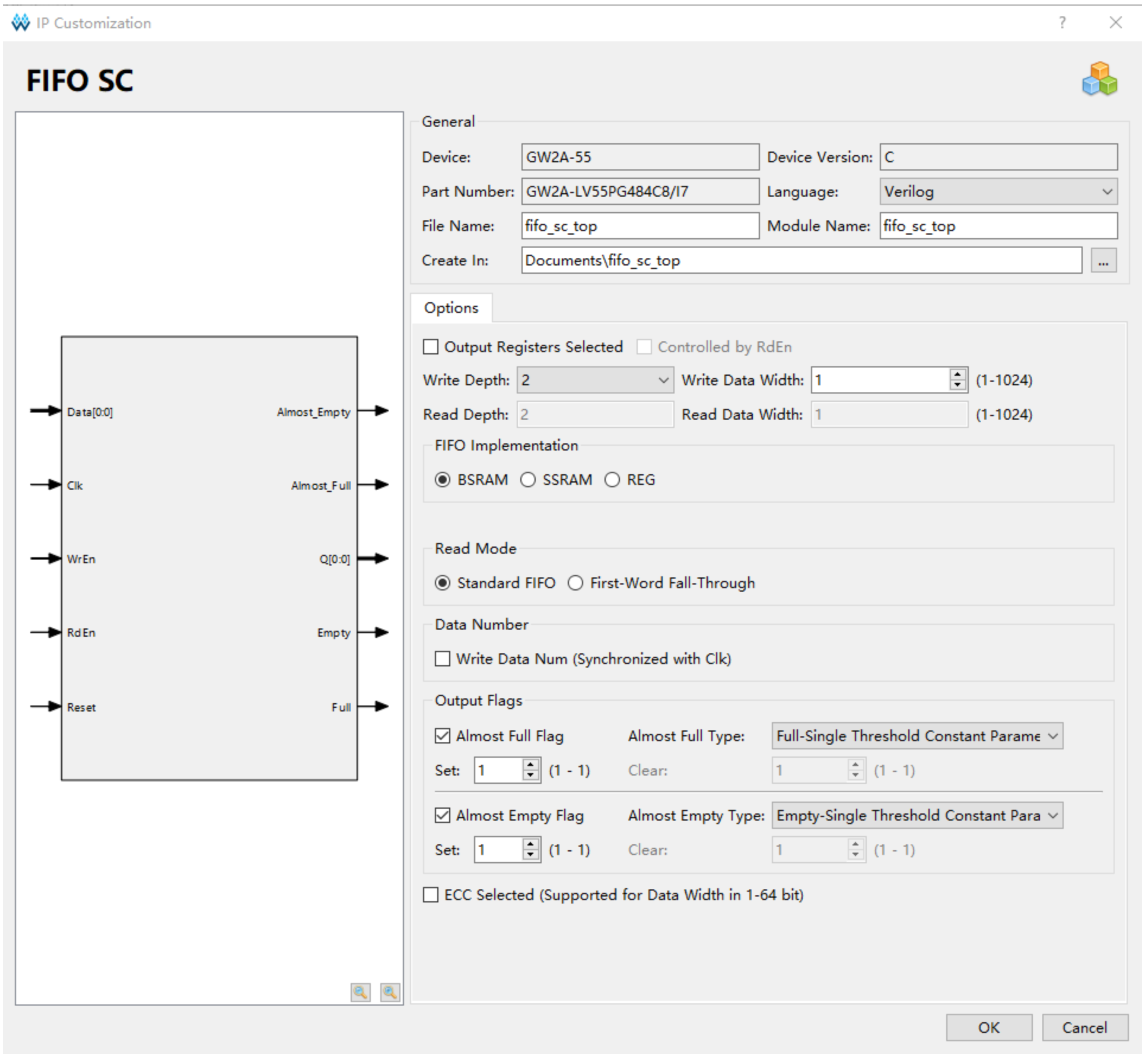


Table 5-2 Gowin FIFO SC IP Parameters

Tab	Parameter	Description	Remarks
General	Device	Select a device	-
	Device Version	Select a version	-

Tab	Parameter		Description	Remarks
	Part Number		Select a Part Number	-
	Language		Select a language	-
	Module Name		FIFO top module name	-
	File Name		FIFO file name	-
	Create In		-	When the FIFO IP model is created, you can choose whether to add it to the current project.
Options	FIFO Implementation	BSRAM	Storage structure is implemented by block SRAM.	-
		SSRAM	Storage structure is implemented by shadow SRAM.	-
		REG	Storage structure is implemented by registers.	-
	Write Depth		Write data depth	FIFO SC IP requires that write data bitwidth and depth is equal to read data bitwidth and depth. There's no need to code Read Depth and Read Data Width because these are equal to Write Depth and Write Data Width.
	Write Data Width		Write data width	
	Read Depth		Read data depth	
	Read Data Width		Read data width	
	Read Mode	Standard FIFO	Standard FIFO	The standard FIFO

Tab	Parameter			Description	Remarks
		First-Word Fall-Through		FWFT FIFO	reads and writes timing according to the standard timing diagram. The FWFT FIFO will write the first data on the output data bus immediately, regardless of whether there is a read enable signal. After the read enable is pulled high, additional data will be outputted in order.
	Data Num	Write Data Num (Synchronized with Clk)		Write data number	Add Rnum output when it is valid.
	Output Flags	Almost Full Flag		Almost full flag enable	Add Almost_Full output when it is valid.
		Almost Full Type	Full-Single Threshold Constant Parameter	When static half-full single threshold is valid, Set is valid.	When Almost Full Flag is valid, users can select one of four options: Full-Single Threshold Constant Parameter, Full-Dual Threshold Constant Parameters, Full-Single Threshold Input Parameter, and Full-Dual Threshold Input Parameters.
			Full-Dual Threshold Constant Parameters	When static half-full dual threshold is valid, Set and Clear are valid.	
			Full-Single Threshold Input Parameter	When Dynamic half-full single threshold input is valid, add AlmostFullTh input.	
			Full-Dual Threshold Input Parameters	When dynamic half-full dual threshold input is valid, add AlmostFullSetTh and AlmostFullClrTh input.	

Tab	Parameter			Description	Remarks
		Set		Set threshold to 1 when half full	● Select single constant threshold, Set is valid. ● Select dual constant threshold, Set and Clear are valid.
		Clear		Set threshold to 0 when half full	
		Almost Empty Flag		Almost empty flag enable	Add Almost_Empty output when it is valid.
		Almost Empty Type	Empty-Single Threshold Constant Parameter	When static half-empty single threshold is valid, Set is valid.	When Almost Empty Flag is valid, users can select one out of four options: Empty -Single Threshold Constant Parameter, Empty-Dual Threshold Constant Parameters, Empty-Single Threshold Input Parameter, and Empty-Dual Threshold Input Parameters.
			Empty-Dual Threshold Constant Parameters	When static half-empty dual threshold is valid, Set and Clear are valid.	
			Empty-Single Threshold Input Parameter	When Dynamic half-empty single threshold input is valid, add AlmostEmptyTh input.	
			Empty-Dual Threshold Input Parameter	When dynamic half-empty dual threshold input is valid, add AlmostEmptySet Th and AlmostEmptyClr Th input.	
		Set		Set threshold to 1 when half empty	Select single constant threshold; Set is valid.
		Clear		Set threshold to 0 when half empty	Select dual constant threshold; Set and Clear are valid.

Tab	Parameter	Description	Remarks
	ECC Selected	ECC	Add ERROR output when ECC is valid.
	Output Register Selected	Output register can be selected.	When valid, data output will be a cycle later. BSRAM output needs to meet the delay from clock to output. When Output Register enable is on, the delay will be met automatically; when Output Register enable is off, the delay from clock to output must be met to avoid sampling metastable data.
	Controlled by RdEn	Output is controlled by RdEn.	When Output Register Selected is valid, it can be selected.

Notes!

- To enable the ECC function, data bitwidth needs to be less than or equal to 64-bit.
- Different output control signals and thresholds can be configured according to the requirements. Table 5-2 shows the FIFO SC IP configuration.

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